

Improving Visual Fidelity of Gaze-Dependent Color Image Presentation Using High-Speed Projection

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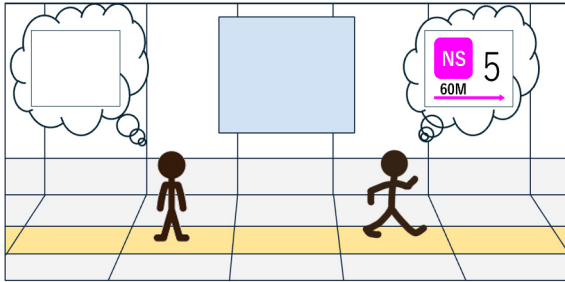


Figure 1: Movement-dependent guidance displays at station.

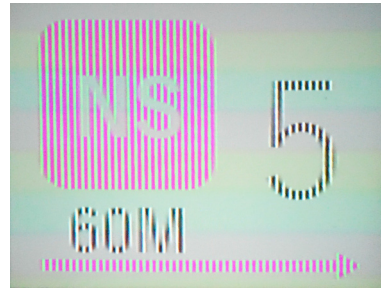


Figure 2: Projected image by using proposed method

ABSTRACT

Large displays and projectors have been widely used in public spaces to present information to multiple viewers simultaneously. While such approaches enable efficient information sharing, they cannot personalize content for individual viewers, and existing solutions using HMDs or camera-based tracking are impractical in public settings due to device scalability and privacy concerns. As an alternative, methods based on high-speed projection have been proposed, in which images become visible only when motion occurs in the viewer's gaze or observation target. In this paper, we propose a colorization method for gaze-dependent image presentation using a high-speed projector that was not achieved by previous approaches, and investigate how variations in the proposed method's parameters affect the fidelity of the presented images.

Index Terms: High-Speed Projection, Projection, Temporal Additive Color Mixing, Augmented Reality

1 INTRODUCTION

Large displays and projectors are widely used as means of presenting information to multiple viewers in various public settings. However, a fundamental limitation of such approaches is that the presented content cannot be personalized for individual observers. To address this issue, several methods have been proposed, including techniques that embed information invisible to the naked eye [1, 4, 9], approaches based on high-speed projection [5, 8], and methods utilizing afterimage effects [2, 3, 6]. Despite their effectiveness, these methods often require additional devices for observation or multiple projectors, which limits their practicality.

One promising approach to overcoming these limitations is a gaze-dependent image presentation method that employs a single high-speed projector and striped pattern images [7]. In this method, visual information is decomposed into a sequence of striped patterns and projected at a high frame rate onto moving objects. When an observer fixates on a target object, the relative motion between

the projected stripes and the retinal image is minimized, allowing the visual system to temporally integrate the patterns into a coherent image. In contrast, when the gaze is not aligned with the object, the projected patterns fail to integrate and remain visually unintelligible. This approach does not require any additional devices beyond the projector itself. However, at present, it is limited to presenting only binary images.

In this paper, we propose a method for improving the visibility of gaze-driven information presentation using colored pattern images. Enabling color presentation allows the proposed approach to convey information that is inherently associated with color semantics, such as route guidance maps and line information in railway stations (Figs. 1 and 2). We investigate how multiple projection conditions—specifically pattern shape, pattern width, and the number of image divisions—affect the reproducibility of the presented images. In addition, we address the unintended appearance of complementary colors introduced by colorization, propose a method to reduce this effect, and quantitatively evaluate its effectiveness.

2 PROPOSED METHOD

This section describes a method for enabling color support in displayed images, which has not been achieved in previous studies, as well as a technique for reducing unintended complementary color artifacts that arise as a result of colorization.

2.1 Colorization Method

In the proposed method, a high-speed projector is used to present color images in response to the observer's eye movements. As illustrated in Fig. 3, the displayed images are generated by decomposing the original image into multiple pattern images that include color inversion in specific regions. Each pixel of the original image, with RGB values (r, g, b) , is classified into either a bright or dark region by comparison with a predefined threshold, and the image is then decomposed into n pattern images.

For pixels classified as bright regions, a uniform value of $(\lfloor \frac{255}{n} \rfloor, \lfloor \frac{255}{n} \rfloor, \lfloor \frac{255}{n} \rfloor)$ is assigned. For pixels in dark regions, pixels that overlap with the pattern are assigned inverted values $(255 - r, 255 - g, 255 - b)$, while the remaining pixels are assigned $(\lfloor \frac{r}{n-1} \rfloor, \lfloor \frac{g}{n-1} \rfloor, \lfloor \frac{b}{n-1} \rfloor)$. The pattern is arranged to match the direction of eye movement; for example, when corresponding to left-to-right eye motion, the pattern is designed to translate from left to right.

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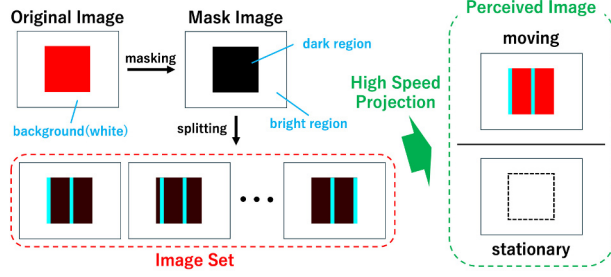


Figure 3: Image decomposition into multiple pattern images and gaze-dependent recombination into visible or invisible images via temporal additive color mixing

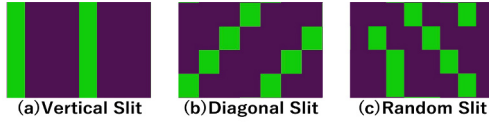


Figure 4: Three types of patterns

Because the human visual system cannot distinguish light stimuli above approximately 60 Hz, sequentially displaying n images at a high refresh rate results in temporal integration, causing each pixel to be perceived as a single color. By exploiting this property, the proposed method preserves the appearance of the original image under static viewing conditions, while the pattern becomes stationary with respect to the visual field only during eye movement, allowing the color image to be perceived.

For comparison, multiple patterns are prepared for the color-filling process, as shown in Fig. 4. All patterns can be adjusted using a common set of parameters: the number of image divisions C and the width W (px) of a single repeating pattern unit.

2.2 Complementary Color Reduction Method (CCR)

In the method described in Sec. 2.1, unintended complementary colors may appear prominently alongside the target color, which degrades image fidelity. This issue is negligible in conventional method for binary image, as its complementary color (white) perceptually blends into the white background. However, for non-black target colors, the complementary colors often exhibit large color differences from both the target color and the background, making them visually salient.

To address this issue, we propose Complementary Color Reduction method (CCR) (Fig. 5). CCR aims to suppress the perceptual saliency of unintended complementary colors while preserving the visibility of the target color. Specifically, we generate an additional image set with a phase shift such that the locations of complementary color regions differ between the two sets. Instead of alternately projecting both sets, we construct a single image set by averaging the pixel values of images with the same index from each set. When projected, complementary color regions in one set are perceptually attenuated by the corresponding target color regions in the other.

This approach effectively reduces the visibility of complementary colors while avoiding overall brightness degradation caused by an increased number of projected images.

3 IMPLEMENTATION

First, a mask image that binarizes the original image into bright and dark regions is generated. This mask is then decomposed into n striped pattern images, to which the color conversion described in Sec. 2.1 is applied. The resulting images are projected using a high-speed projector (STAR-07, ViALUX) controlled in a Windows environment. During the experiment, a separate projector controlled by a macOS-based PC was used to display a gaze-guidance point.

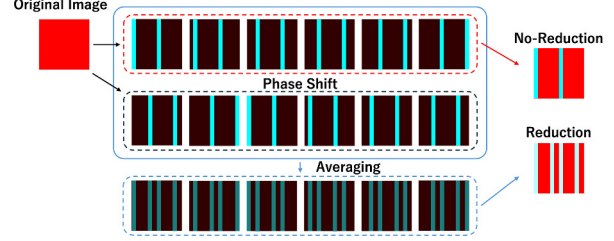


Figure 5: Flow of CCR process

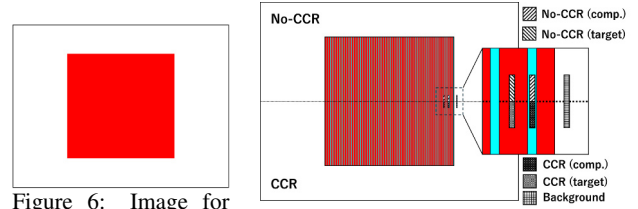


Figure 6: Image for CCR evaluation

Figure 7: Representative regions used for evaluation

Participants observed the projected images while their head position was stabilized using a fixed face stabilizer. The projection screen was placed 135cm in front of the participant's viewpoint, and the images were projected onto the screen with a size of 45cm×60cm.

4 EVALUATION(ASSESSMENT OF THE OCCURRENCE RATIO OF COMPLEMENTARY COLOR REGIONS)

4.1 Objective

The objective of this evaluation is to verify that the CCR decreases the visual saliency of complementary colors in the displayed images.

4.2 Hypothesis

By applying CCR, the colors of complementary regions produced by the colorization method can be made less visually conspicuous.

4.3 Evaluation Procedure

To evaluate the performance of CCR, we used an image in which a square with color $(r, g, b) = (255, 0, 0)$ was placed at the center, as shown in Fig. 6. The same input image is processed using both CCR and the colorization method incorporating CCR. The resulting images are vertically divided and combined into a single image. For CCR, an image set obtained by integrating two image sets into one is used.

The combined image is projected and captured using a camera. Pixel values corresponding to the target color region, complementary color region, and background are numerically compared and evaluated. For the evaluation, a single frame is extracted from the captured video, and rectangular regions of identical shape are selected from both the target color region and the complementary color region (Fig. 7).

Color differences are evaluated using the CIEDE2000 metric. Specifically, color differences between the target color and complementary color regions, between the background and complementary color regions, and between the background and target color regions are computed and compared with and without CCR.

4.4 Results

For both the target color–complementary color and background–complementary color pairs, the color differences were significantly reduced compared to those obtained without CCR. In contrast, for the target color–background pair, little change in color

Table 1: Average RGB pixel values of representative regions and the relative contribution of each channel to the total RGB intensity (Red as the target color)

Region (CCR)	Pixel values (Ratio to the sum of pixel values)		
–	<i>R</i>	<i>G</i>	<i>B</i>
Comp. (No)	81.36 (0.16)	206.95 (0.42)	209.29 (0.42)
Comp. (Yes)	78.91 (0.23)	136.98 (0.39)	134.28 (0.38)
Target (No)	115.99 (0.56)	52.54 (0.26)	37.51 (0.18)
Target (Yes)	113.80 (0.48)	67.00 (0.29)	54.38 (0.23)
Back. (–)	83.79 (0.28)	104.85 (0.36)	106.49 (0.36)

Table 2: Changes in CIEDE2000 color differences due to CCR

Compared regions		No CCR	CCR
Complementary – Target		63.33	41.36
Complementary – Background		33.59	14.00
Target – Background		33.48	30.51

difference was observed regardless of whether CCR was applied (Tabs. 1 and 2).

4.5 Discussion

Based on the evaluation results, CCR exhibits two main effects. First, the complementary color regions become closer in color to the target color regions, which reduces their visual saliency. In addition, these regions also shift toward the background color, allowing them to blend more easily into the background.

Together, these effects indicate that CCR effectively suppresses the conspicuity of complementary color regions, thereby supporting the hypothesis. On the other hand, this method doubles the area occupied by complementary color regions, resulting in a trade-off between their spatial extent and visual saliency in the proposed approach.

5 EXPERIMENT (EVALUATION OF FIDELITY)

5.1 Objective

This experiment examines how pattern width (W), pattern shape, and the number of image divisions (C) affect image fidelity. Fidelity is defined as the agreement between simulation results and images perceived by human observers, and is evaluated in terms of shape fidelity and color fidelity. Shape fidelity represents the accuracy of perceived spatial structure relative to the simulation, while color fidelity represents the accuracy of perceived color appearance.

5.2 Hypotheses

We formulate the following hypotheses:

- H.1** The width parameter W affects image fidelity; narrower widths improve shape fidelity, whereas wider widths improve color fidelity.
- H.2** The pattern shape parameter affects image fidelity; patterns with more feature points improve both color and shape fidelity.
- H.3** The division parameter C affects image fidelity; increasing the number of divisions improves color fidelity while degrading shape fidelity.

5.3 Apparatus

The experiment was conducted in a dark room using a high-speed projector for image presentation and an additional projector for projecting gaze-guidance points. The high-speed projector was controlled by PC (macOS Tahoe), and the gaze-guidance projector was controlled by a separate PC (Windows 11) (Fig. 8).

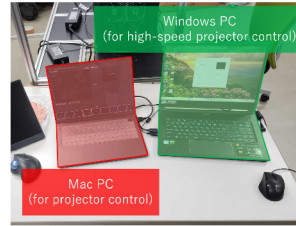


Figure 8: Control PC

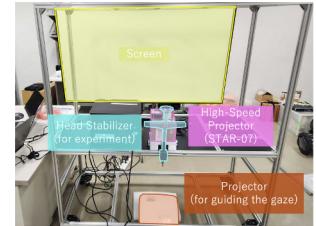


Figure 9: Projectors and Screen

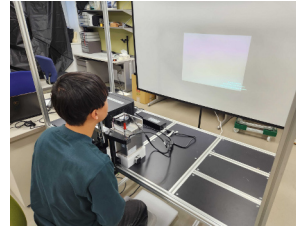


Figure 10: Scene from the experiment

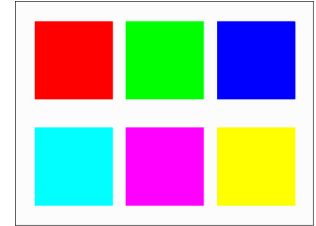


Figure 11: Image for color selection

The presented images were projected onto a screen located 135 cm from the participant, with a display size of 60 cm × 45 cm. Participants observed the images with their chin fixed on a chin rest mounted on a stand at a fixed position relative to the screen. They were instructed to keep their head still and to move only their eyes during observation (Figs. 9 and 10).

5.4 Experimental Procedure

Twelve participants (9 males, 3 females; age 21–34 years, mean 23.8) with normal or corrected-to-normal vision and visual acuity of 0.7 or higher participated in the experiment. Four participants wore glasses and three wore contact lenses. Each participant received 1,000 JPY as compensation.

A projector separate from the high-speed projector was used to display a gaze-guidance point on the projection screen. The point was either stationary or moved horizontally at a constant velocity adjusted for each projected pattern. Prior to the experiment, participants completed a training session to ensure reliable gaze tracking at all tested speeds.

Participants first selected the most visually discernible color from six candidates (red, green, blue, cyan, magenta, and yellow; Fig. 11). Six color-filled squares were projected together with the gaze-guidance point using a vertical slit pattern ($W = 3$ px, $C = 6$). The color arrangement was undisclosed; each region was labeled, and participants reported the number corresponding to the most visible square. The selected color was used throughout the experiment.

For fidelity evaluation, participants compared the perceived appearance of the projected images with simulation results. The simulation represented the image perceived during gaze movement and was generated by gradually shifting the image horizontally while accumulating pixel values from each image in the imageset. The resulting simulated image was statically projected without image decomposition and served as a stable reference for the expected perceptual appearance during gaze shifts.

Three variations were prepared for each parameter (W , pattern shape, and C) and arranged vertically within a single image for comparison (Fig. 12). Participants compared the simulated results with the perceived appearance of the projected images. The presentation order of parameter groups was randomized across participants.

Pattern widths of 1,px (minimum configurable), 3,px (identified as optimal in a pilot study), and 6,px (corresponding to the maximum gaze-tracking speed) were tested. Pattern shapes included vertical slits used in prior work; diagonal slits with an increased

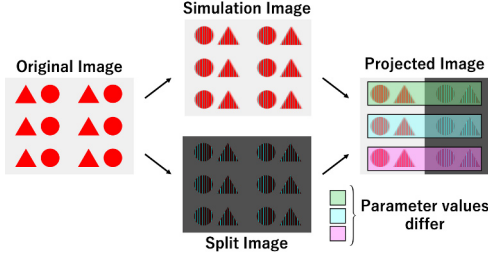


Figure 12: Projected image in experiment

Table 3: Combinations of parameters used in the experiment

Parameter	W [px]	Pattern	C
Width W [px]	1	Vertical slit	6
	3	Vertical slit	6
	6	Vertical slit	6
Pattern shape	3	Vertical slit	6
	3	Diagonal slit	6
	3	Random slit	6
Division count C	3	Vertical slit	4
	3	Vertical slit	6
	3	Vertical slit	12

number of feature points, defined as corners and edges in the pattern outline, while preserving regularity; and random slits derived from the diagonal patterns with further reduced regularity. The number of image divisions was set to 4, 6, and 12 based on pilot study results and brightness constraints (Tab. 3).

This study was approved by the Ethics Committee of Institute of Science Tokyo.

5.5 Results

5.5.1 Shape Fidelity

We used a Friedman test to examine the effects of each parameter, with the significance level set to 5% ($\alpha = 0.05$). Significant effects were observed for pattern width and pattern shape (width: $p = 0.002$, shape: $p = 0.005$), whereas no significant effect was found for the number of image divisions ($p = 0.105$).

Post-hoc analysis using the Wilcoxon signed-rank test with Bonferroni correction revealed that the 3 px width condition differed significantly from both the 1 px and 6 px conditions. In addition, a significant difference was observed between the vertical slit pattern and the random slit pattern (Tab. 4 and Fig. 13).

5.5.2 Color Fidelity

A Friedman test was conducted with the significance level set to 5% ($\alpha = 0.05$). Significant effects were observed for width and pattern shape (width: $p < 0.001$, shape: $p = 0.011$), while no significant effect was found for the number of image divisions ($p = 0.074$). For parameters showing significant effects, post-hoc analysis using the Wilcoxon signed-rank test with Bonferroni correction indicated that the 1 px width condition differed significantly from both the 3 px and 6 px conditions. (Tab. 5 and Fig. 14).

5.6 Discussion

The results clarify how each parameter affects image fidelity.

For the width parameter (W), Hypothesis H.1 was partially supported. Although W significantly influenced both shape and color fidelity, the relationship was non-monotonic: shape fidelity peaked at intermediate widths, while narrow widths significantly degraded color fidelity without proportional improvement at wider settings.

Table 4: Statistical comparison results for shape fidelity

Parameter	χ^2 (df=2)	p	Significant pairs
W	12.60	.002	1–3, 3–6
Pattern shape	10.75	.005	Vertical–Random
C	4.50	.105	None

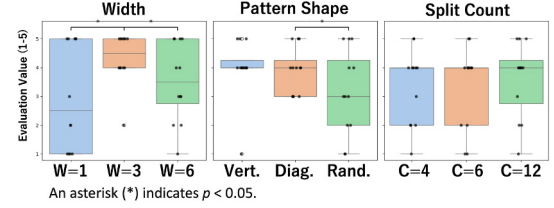


Figure 13: Shape Fidelity

Table 5: Statistical comparison results for color fidelity

Parameter	χ^2 (df=2)	p	Significant pairs
W	15.50	.000	1–3, 1–6
Pattern shape	9.00	.011	None
C	5.20	.074	None

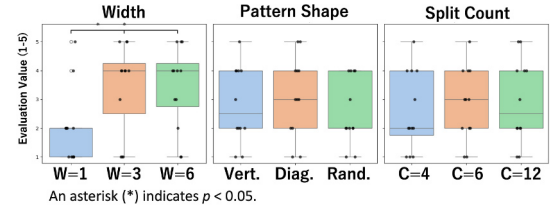


Figure 14: Color Fidelity

This indicates an optimal range of W that balances both fidelities rather than a simple trade-off.

Hypothesis H.2 was not supported. Contrary to expectations, the conventional vertical slit pattern achieved higher shape and color fidelity than feature-rich patterns. Although a significant difference in color fidelity was found between the two feature-dense patterns, no consistent relationship with feature-point density was observed, suggesting that pattern orientation and spatial continuity are more influential.

Hypothesis H.3 was rejected, as no significant differences were observed across the tested values of the division parameter C , indicating robustness within the evaluated range.

Overall, these results suggest that perceptual image reconstruction depends on balanced parameter selection and pattern structure rather than monotonic parameter scaling.

6 CONCLUSION AND DISCUSSION

This paper proposed a gaze-shift-contingent color image presentation method using high-speed projection and introduced Complementary Color Reduction (CCR) to suppress the saliency of unintended complementary colors. Experiments showed that image fidelity depends on balanced parameter selection, with intermediate pattern widths and conventional vertical slit patterns achieving higher color and shape fidelity. These results provide practical guidelines for device-free, gaze-responsive personalized information presentation in public spaces.

Limitations include narrow acceptable range of gaze-shift speed, reduced color gradation and brightness with increasing image count, and the requirement for high-refresh-rate projectors. Future work is expected to address gaze-speed robustness, perceptually informed gradation reduction, and multi-projector approaches to improve brightness and effective refresh rates.

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